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Carrier measurement device, carrier measurement method, and carrier management method

Abstract

A carrier measuring device includes a rotary table, a table drive motor, an upper thickness sensor, a lower thickness sensor and a slide unit. The rotary table includes a carrier receiver configured to horizontally house a carrier formed with a hole in which a semiconductor wafer is held, the hole being eccentric with respect to the carrier. The table drive motor rotates the rotary table around a center axis thereof as a rotation axis. The upper thickness sensor and the lower thickness sensor are positioned above and below the carrier, respectively, and measure a thickness of the carrier in a non-contact manner. The slide unit slides the rotary table in a horizontal direction. The carrier receiver is formed to be capable of housing the carrier in a manner that a center of the hole coincides with a center of the rotary table.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3735534	12/1972	Mayo	451/6	B24B 13/00
5234499	12/1992	Sasaki	118/52	H01L 21/6715
6544103	12/2002	Korovin	N/A	N/A
10391603	12/2018	Yamaki	N/A	B24B 37/005
10569381	12/2019	Yoshida	N/A	B24B 37/32
10605583	12/2019	Honda	N/A	G01G 19/083
10622232	12/2019	Ha	N/A	H01L 21/67115
10699926	12/2019	Plihal	N/A	G06T 7/001
11171056	12/2020	Yamamoto	N/A	H01L 21/67288
11387726	12/2021	Sakai	N/A	H02K 41/031
11450523	12/2021	Tanoue	N/A	B23K 26/032
11598007	12/2022	Akashi	N/A	C23C 16/46
12183642	12/2023	Nabeya	N/A	H01L 22/26
2017/0069502	12/2016	Sato et al.	N/A	N/A
2021/0164769	12/2020	Jung	N/A	N/A
2024/0238934	12/2023	Ota	N/A	B24B 37/005
2024/0353344	12/2023	Kinoshita	N/A	H01L 22/12
2025/0046646	12/2024	Taguchi	N/A	H01L 21/68771
2025/0051920	12/2024	Fukushi	N/A	H01L 21/68771

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC

2012-145439	12/2011	JP	N/A
5732858	12/2014	JP	N/A
2015-174168	12/2014	JP	N/A
6056793	12/2016	JP	N/A
2017-159382	12/2016	JP	N/A
2019-509465	12/2018	JP	N/A
6578442	12/2018	JP	N/A
2019-186490	12/2018	JP	N/A
2020-49612	12/2019	JP	N/A
10-2000-0003078	12/1999	KR	N/A
10-2013-0084861	12/2012	KR	N/A
10-2016-0133437	12/2015	KR	N/A
10-2017-0104925	12/2016	KR	N/A
10-1856875	12/2017	KR	N/A

OTHER PUBLICATIONS

International Search Report issued in WIPO Patent Application No. PCT/JP2021/036539, dated Dec. 21, 2021, along with an English translation thereof. cited by applicant

Written Opinion issued in WIPO Patent Application No. PCT/JP2021/036539, dated Dec. 21, 2021. cited by applicant

Office Action issued in Corresponding KR Patent Application No. 10-2023-7012110, dated Jan. 20, 2025, along with an English translation thereof. cited by applicant

International Preliminary Report on Patentability issued in Corresponding International Application No. PCT/JP2021/036539, dated Apr. 13, 2023, along with an English translation thereof. cited by applicant

Office Action issued in Corresponding KR Patent Application No. 10-2023-7012110, dated Sep. 23, 2024, along with an English translation thereof. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a carrier measuring device and a carrier measurement method for measuring a thickness and a shape of carriers each having a hole for holding a disc-shaped semiconductor wafer, and a carrier management method for managing carriers on a basis of measurement results obtained by measuring a thickness and a shape of the carriers.

BACKGROUND ART

(2) In a manufacturing process of semiconductor wafers used as substrates of semiconductor devices, in order to attain high flatness and low surface roughness of the semiconductor wafers, both sides of the semiconductor wafers are simultaneously polished (i.e., double-side polishing of each semiconductor wafer is performed). Specifically, a plurality of disc-shaped carriers respectively holding the semiconductor wafers in circular holes formed therein are mounted in a double-side polishing device, and both sides of each semiconductor wafer are simultaneously polished while the semiconductor wafers are sandwiched between a pair of platens each attached with a polishing pad.

(3) A region of the semiconductor wafer in which the semiconductor devices are formed has recently become larger year by year toward an outer circumferential portion from a central portion of the semiconductor wafer. Thus, higher flatness has been required in not only the central portion but also the outer circumferential portion of the semiconductor wafer. In order to polish the outer circumferential portion of the semiconductor wafer (hereinafter abbreviated as a “wafer outer circumferential portion”) so that the wafer outer circumferential portion has high flatness, it is important that the carrier having the hole for holding the semiconductor wafer should have a highly flat portion surrounding the hole (hereinafter abbreviated as a “hole surrounding portion”).
(4) This is because low flatness, for instance, a thin part (occasionally also referred to as a “roll-off”) of the hole surrounding portion of the carrier causes the polishing pad for double-side polishing to enter the thin part and excessively polish to thin the wafer outer circumferential portion since the wafer outer circumferential portion is in contact with the hole surrounding portion (this phenomenon is also referred to as “roll-off”).
(5) Since the plurality of carriers holding the respective semiconductor wafers are mounted in the double-side polishing device for polishing, it is required that not only the individual carriers should have high flatness but also the carriers mounted together in the double-side polishing device in performing one batch of double-side polishing should have constant thickness and shape.
(6) In a case where the thickness and shape of the carriers are varied among the carriers used in one batch of double-side polishing, the polishing pad is intensively pressed against the carrier having roll-off in the hole surrounding portion, resulting in a variation in shape among the semiconductor wafers in the batch to cause a variation in quality among the semiconductor wafers polished in the batch.
(7) Therefore, as preparation for double-side polishing of the semiconductor wafers, it is important to accurately measure the thickness and shape of each carrier (particularly the hole surrounding portion thereof) and select carriers to be used in one batch from the carriers having less variance in thickness and shape.
(8) As described in, for instance, Patent Literature 1, a conventional measuring device configured to measure the thickness of the carrier includes a table on which the carrier is placeable, a reference rod, a contact sensor and a controller.
(9) The controller, while the carrier is not placed on the table, brings the reference rod and the contact sensor into contact with the table and determines a reference value at a point where the contact sensor contacts with the table. Next, the controller, while the carrier is placed on the table, brings the reference rod and the contact sensor respectively into contact with an upper surface of the carrier and the table and determines a measurement value at a point where the contact sensor contacts with the table. The controller calculates the thickness of the carrier by using a difference between the reference value and the measurement value.
(10) As described in, for instance, Patent Literature 2, another conventional measuring device for the thickness of the carrier includes a first table, a second table, an upper sensor, a lower sensor and a sensor drive unit.
(11) The first table, which is disc-shaped, is disposed in a rotatable manner and supports a central portion of the carrier. The second table, which is shaped in a ring plate, is disposed outside the first table in a rotatable manner and supports an outer circumferential portion of the carrier.
(12) The upper sensor and the lower sensor, which are respectively attached to an end of an upper arm portion and an end of a lower arm portion of a C-shaped support member, calculate the thickness of the carrier. The sensor drive unit moves the upper sensor and the lower sensor by rotating the support member to position the upper sensor and the lower sensor above and below the carrier, respectively.
(13) The second table is rotated with the outer circumferential portion of the carrier being supported by the second table, and the upper sensor and the lower sensor measure a thickness of an inner circumferential portion of the carrier. Moreover, the first table is rotated with the inner

circumferential portion of the carrier being supported by the first table, and the upper sensor and the lower sensor measure a thickness of the outer circumferential portion of the carrier.
(14) Further, Patent Literature 3 describes that a flatness distribution of the portion surrounding the hole formed in the carrier is measured by using a laser displacement sensor.
CITATION LIST
Patent Literature(s)
(15) Patent Literature 1: JP Patent No. 5732858 (claim 1) Patent Literature 2: JP Patent No. 6578442 (paragraphs 0033 to 0039, 0045 to 0049 and 0063 to 0072 and FIGS. 7, 8 and 9a to 9i) Patent Literature 3: JP Patent No. 6056793 (paragraph 0052)
SUMMARY OF THE INVENTION
Problem(s) to be Solved by the Invention
(16) In the conventional art described in Patent Literature 1, measurement positions are selected one by one and the thickness of the carrier is measured at each position. Thus, a large amount of time and effort is required to measure the thickness of the carrier along the entire hole surrounding portion and the carrier having roll-off in the hole surrounding portion cannot be easily and quickly detected.
(17) In order to hold the wafer so that the wafer is eccentric with respect to the carrier, the carrier may be formed with a hole whose center is offset from a center of the carrier. However, in the conventional art described in Patent Literature 2, in the measurement, the carrier is rotated around a center axis thereof and the upper sensor and the lower sensor are moved by being rotated by the sensor drive unit to be positioned above and below the carrier, respectively. Thus, in the conventional art described in Patent Literature 2, the thickness of the carrier along the entire hole surrounding portion that surrounds the eccentric hole cannot be measured, so that the carrier having roll-off in the hole surrounding portion cannot be detected.
(18) In this regard, Patent Literature 3, which describes in paragraph 0052 that the flatness distribution of the portion surrounding the hole formed in the carrier is measured by using a laser displacement sensor, fails to describe any specific configuration of a measuring device used for the measurement.
(19) Therefore, there has been desired a configuration of accurately measuring the thickness and shape of each carrier (particularly the hole surrounding portion that surrounds the eccentric hole thereof), thereby enabling selection of carriers to be used in one batch from the carriers having less variance in thickness and shape.
(20) An object of the invention is to provide a carrier measuring device, a carrier measurement method and a carrier management method capable of accurately measuring a thickness and a shape of each of carriers (particularly a hole surrounding portion that surrounds an eccentric hole thereof) and enabling selection of carriers to be used in one batch from the carriers having less variance in thickness and shape.
Means for Solving the Problem(s)
(21) According to an aspect of the invention, a carrier measuring device configured to measure a thickness of a carrier having a hole in which a disc-shaped semiconductor wafer is to be held, the hole being eccentric with respect to the carrier, includes: a rotary table including a carrier receiver configured to horizontally house the carrier; a table drive motor configured to rotate the rotary table around a center axis thereof as a rotation axis; an upper thickness sensor and a lower thickness sensor configured to be positioned above and below the carrier, respectively, and measure the thickness of the carrier in a non-contact manner; and a slide unit configured to slide, in a horizontal direction, one or both of: the rotary table; and the upper thickness sensor and the lower thickness sensor, in which the carrier receiver is formed to be capable of housing the carrier in a manner that a center of the hole of the carrier coincides with a center of the rotary table.
(22) The carrier measuring device according to the aspect of the invention further includes a sensor frame attached with the upper thickness sensor and the lower thickness sensor between which the

rotary table is capable of passing through in the horizontal direction.
(23) In the carrier measuring device according to the aspect of the invention, the upper thickness sensor and the lower thickness sensor are attached to the sensor frame in a manner to be respectively positioned above and below a straight line passing through the center of the rotary table and a center of the carrier receiver, and the slide unit slides one or both of the rotary table and the sensor frame in the horizontal direction parallel to the straight line.
(24) According to another aspect of the invention, a carrier measurement method for measuring, by using the above carrier measuring device, a thickness of a disc-shaped carrier having a hole in which a disc-shaped semiconductor wafer is to be held, the hole being eccentric with respect to the carrier, includes: first measuring the thickness of the carrier at a predetermined point while sliding, in the horizontal direction, one or both of: the rotary table in which the carrier is housed; and the upper thickness sensor and the lower thickness sensor; and second measuring a thickness of a hole surrounding portion that surrounds the hole of the carrier while rotating the rotary table in which the carrier is housed.
(25) According to still another aspect of the invention, a carrier management method for managing, on a basis of measurement results obtained through measurement using the above carrier measuring device, a plurality of disc-shaped carriers each having a hole in which a disc-shaped semiconductor wafer is to be held, the hole being eccentric with respect to the corresponding carrier, includes: first measuring the thickness of each of the carriers at a predetermined point while sliding, in the horizontal direction, one or both of: the rotary table in which the carrier is housed; and the upper thickness sensor and the lower thickness sensor; classifying the carriers falling within a predetermined range of the thickness into the same class on a basis of the measurement results in the first measuring; selecting a predetermined number of carriers from the carriers belonging to the same class to determine a carrier set to be used in the same batch in double-side polishing of the semiconductor wafers; second measuring a thickness of a hole surrounding portion that surrounds the hole of each of the carriers of the carrier set while rotating the rotary table in which the carriers are housed; and ranking the carrier set on a basis of the measurement results in the first and second measuring according to a specification required for the semiconductor wafers.
(26) In the carrier management method according to the aspect of the invention, in the ranking, a roll-off amount of the hole surrounding portion of each of the carriers is calculated on a basis of the measurement results in the second measuring and the carrier set is ranked on a basis of the measurement results in the first measuring and the roll-off amounts of the hole surrounding portions.
(27) According to the above aspects of the invention, the thickness and shape of each carrier (particularly the hole surrounding portion that surrounds the eccentric hole thereof) can be accurately measured, so that carriers to be used in one batch can be selected from the carriers having less variance in thickness and shape.

Description

BRIEF DESCRIPTION OF DRAWING(S)
(1) FIG. **1** is a top view showing an example of a schematic structure of a carrier measuring device according to an exemplary embodiment of the invention.
(2) FIG. **2** is a right-side view of the carrier measuring device shown in FIG. **1**.
(3) FIG. **3** is a block diagram showing an example of a schematic structure of a control system in the carrier measuring device shown in FIG. **1**.
(4) FIG. **4**A is a top view showing an example of an external configuration of a carrier.
(5) FIG. **4**B illustrates an enlarged view of a part of a hole surrounding portion of the carrier and also illustrates a graph showing an example of a relationship between a distance from a hole edge

and a thickness of the carrier.
(6) FIG. **5** is a flowchart showing an example of a carrier management method for standard semiconductor wafers according to an exemplary embodiment of the invention.
(7) FIG. **6** is a flowchart showing an example of a carrier management method for high-precision semiconductor wafers according to the exemplary embodiment of the invention.
(8) FIG. **7** is a diagram showing an example of a relationship between a maximum roll-off amount of hole surrounding portions of respective carriers and a roll-off amount of the corresponding wafer outer circumferential portions.
DESCRIPTION OF EMBODIMENT(S)
(9) Exemplary embodiments of the invention are described below with reference to the attached drawings.
(10) Structure of Cartier Measuring Device
(11) FIGS. **1** and **2** are respectively a top view and a right-side view showing an example of a schematic structure of a carrier measuring device **1** according to an exemplary embodiment of the invention. FIG. **3** is a block diagram showing an example of a schematic structure of a control system in the carrier measuring device **1**.
(12) The carrier measuring device **1** includes a platform **11**, which has a frame structure shaped in a rectangular parallelepiped as a whole. A rectangular plate-shaped base **31** is fixed on the platform **11**.
(13) A guide rail **32** is fixed on an upper surface of the base **31** at each of sides in a longitudinal direction thereof. A rectangular plate-shaped stage **12** is placed on the guide rails **32** in a slidable manner.
(14) A ball screw **33** is disposed in a middle in a width direction (an X-axis direction shown in FIG. **1**) of the base **31**.
(15) As shown in FIG. **2**, an Y-axis motor **34** is attached to one end of the ball screw **33**, specifically to a rear end of the ball screw **33** in a Y-axis direction. Meanwhile, a lower portion **12**A of the stage **12** is attached to a position close to the other end of the ball screw **33**, specifically to a position close to a front end of the ball screw **33** in the Y-axis direction.
(16) This arrangement enables, by rotational drive of the Y-axis motor **34**, the stage **12** to slide on the guide rails **32** in front and back (i.e., positive and negative) directions of the Y axis via the ball screw **33**. Here, a slide unit **13** includes the base **31**, the pair of guide rails **32**, the ball screw **33** and the Y-axis motor **34**.
(17) Hereinafter, a longitudinal direction of the base **31** from a back toward a front of the carrier measuring device **1** is defined as a Y-axis positive direction and a longitudinal direction of the base **31** from the front toward the back of the carrier measuring device **1** is defined as a Y-axis negative direction.
(18) A table receiver **22**A is formed in an upper surface of the stage **12**.
(19) A disc-shaped rotary table **23** is housed in the table receiver **22**A in a rotatable manner.
(20) A carrier receiver **23**A, which is in a form of a circular recess and has a slightly larger diameter than a later-described carrier **21**, is formed in an upper surface of the rotary table **23**. The carrier **21**, which is disc-shaped, is horizontally housed in the carrier receiver **23**A.
(21) An outer circumferential portion of the carrier **21** is provided with outer circumferential teeth **21**B at a predetermined pitch. The carrier **21** may include a body made from a metal material such as stainless steel or titanium and a ring-shaped resin inserter disposed along an inner wall of the carrier **21**, the inner wall defining a hole **21**A for holding a semiconductor wafer. Alternatively, an entirety of the carrier **21** may be formed from a resin. When the carrier **21** is made from a resin, the resin to be used is preferably a composite of an epoxy resin and a glass fiber.
(22) The rotary table **23** is rotationally driven by a table drive motor **24**.
(23) A sensor frame **14**, which is flat O-shaped (i.e., hollow rectangular) in a front view thereof, is fixed on a central portion of an upper surface of the platform **11** across a width direction of the

platform **11**. The stage **12** can pass through an opening **14**A of the sensor frame **14** in the Y-axis direction.
(24) An upper thickness sensor **41** and a lower thickness sensor **42** are each attached to the sensor frame **14** at a middle in a longitudinal direction on a front thereof, the upper thickness sensor **41** and the lower thickness sensor **42** being at a predetermined distance from each other in a vertical direction. The upper thickness sensor **41** and the lower thickness sensor **42** are each, for instance, a spectral interference laser displacement sensor. The upper thickness sensor **41** and the lower thickness sensor **42** can measure a thickness along an entire circumference of the hole surrounding portion **21**C defining the hole **21**A formed in the carrier **21** at one point per predetermined central angle of the hole **21**A specified by an operator. Further, in a measurable area of the hole surrounding portion **21**C, measurement is performed at innermost points and then points gradually radially outward, enabling expansion of a measurement circle.
(25) In measuring the thickness of the hole surrounding portion **21**C, the table drive motor **24** horizontally rotates the rotary table **23** around a center axis vertically passing through a center C**1** of the rotary table **23** as a rotation axis by the central angle specified by the operator. In accordance with rotation of the rotary table **23**, the upper thickness sensor **41** and the lower thickness sensor **42** measure the thickness of the hole surrounding portion **21**C along its entire circumference defining the hole **21**A formed in the carrier **21** at one point per predetermined central angle of the hole **21**A specified by the operator. The upper thickness sensor **41** and the lower thickness sensor **42** can measure the thickness of the hold surrounding portion **21**C, for instance, at a total of 360 points (i.e., at one point per degree of the central angle of the hole **21**A).
(26) The platform **11** is installed with a computer **43** configured to control each of components of the device. The computer **43** includes: a controller **51** including a Central Processing Unit (CPU) and the like; and a storage **52** including a recording circuit such as a memory or a hard disk (see FIG. **3**).
(27) The controller **51** reads and executes various programs (e.g., a carrier measurement program and a carrier management program) stored in the storage **52**. The storage **52** stores the various programs (e.g., the carrier measurement program and the carrier management program) and various data used for executing the various programs.
(28) As shown in FIG. **1**, a center C**2** of the carrier receiver **23**A is eccentric with respect to the center C**1** of the rotary table **23** toward the back of the carrier measuring device **1** (i.e., in the Y negative direction). In the exemplary embodiment, a radial distance from the center C**2** of the carrier receiver **23**A to the center C**1** of the rotary table **23** is defined as an eccentricity amount of the carrier receiver **23**A with respect to the rotary table **23**.
(29) In the exemplary embodiment, when a straight line L passing through the center C**2** of the carrier receiver **23**A and the center C**1** of the rotary table **23** is parallel to the Y direction as shown in FIG. **1**, the center C**2** of the carrier receiver **23**A is positioned close to an outer circumference of the rotary table **23** with respect to the center C**1** of the rotary table **23** (i.e., close to either the front or back of the carrier measuring device **1**). A direction indicating this position of the center C**2** of the carrier receiver **23**A with respect to the center C**1** of the rotary table **23** is defined as an eccentricity direction of the carrier receiver **23**A. In the example of FIG. **1**, since the center C**2** of the carrier receiver **23**A is positioned close to the back of the carrier measuring device **1** with respect to the center C**1** of the rotary table **23**, the eccentricity direction of the carrier receiver **23**A is the Y negative direction.
(30) In the example shown in FIG. **1**, the carrier **21** is housed in the carrier receiver **23**A such that a center C**4** of the hole **21**A formed in the carrier **21** is eccentric with respect to a center C**3** of the carrier **21** toward the front of the carrier measuring device **1** (i.e., in the Y positive direction). That is, an eccentricity direction of the hole **21**A is the Y positive direction. It should be noted that the center C**3** of the carrier **21** coincides with the center C**2** of the carrier receiver **23**A.
(31) The reason why the carrier receiver **23**A is formed as shown in FIG. **1** and the carrier **21** is

housed in the carrier receiver **23**A in the direction shown in FIG. **1** is as follows.
(32) A conventionally known device for simultaneously polishing both sides of a plurality of semiconductor wafers is a double-side polishing device using a planetary gear mechanism. The double-side polishing device includes: a rotary platen including an upper platen and a lower platen; a sun gear provided in a center of the rotary platen; an internal gear provided at an outer circumferential portion of the rotary platen; and a plurality of carriers provided between the upper platen and the lower platen, an outer circumference of each carrier being formed with outer circumferential teeth at a predetermined pitch.
(33) In the double-side polishing device, each carrier rotates between the upper platen and the lower platen by the outer circumferential teeth of the carrier engaging with each of the sun gear and the internal gear, whereby planetary movement of the carrier is performed in which the carrier revolves around the sun gear while rotating around a center axis thereof. In the planetary movement, when a center of the carrier coincides with a center of a hole formed in the carrier, both sides of the semiconductor wafer housed in the hole are always polished by the same portion of a polishing pad, so that not only a polishing efficiency is low but also desired flatness of the semiconductor wafer is not obtainable.
(34) In order to enhance the polishing efficiency of the semiconductor wafer and uniformity in polishing of the semiconductor wafer, the center of the carrier and the center of the hole are made not to coincide with each other, i.e., the hole is made eccentric with respect to the carrier.
(35) By simply rotating the rotary table **23** in which the carrier **21** formed with the hole **21**A that is eccentric with respect to the carrier **21** for the above reason is housed in the carrier receiver **23**A whose center coincides with the center C**1** of the rotary table **23**, the thickness of the hole surrounding portion **21**C defining the eccentric hole **21**A cannot be measured along the entire circumference thereof.
(36) Thus, in the exemplary embodiment, an eccentricity amount of the hole **21**A with respect to the carrier **21** is set the same as the eccentricity amount of the carrier receiver **23**A with respect to the rotary table **23** and the eccentricity direction of the hole **21**A (i.e., Y positive direction) is set opposite by 180 degrees to the eccentricity direction of the carrier receiver **23**A (i.e., Y negative direction), as shown in FIG. **1**. In this arrangement, when the carrier **21** is housed in the carrier receiver **23**A of the rotary table **23**, eccentricity of the hole **21**A with respect to the carrier **21** is offset by eccentricity of the carrier receiver **23**A with respect to the rotary table **23**. Therefore, since the center C**4** of the hole **21**A coincides with the center C**1** of the rotary table **23**, by rotating the rotary table **23** around the center axis thereof as a rotation axis with the upper thickness sensor **41** and the lower thickness sensor **42** (see FIGS. **1** and **2**) remaining fixed in alignment in a certain position, the thickness of the hole surrounding portion **21**C that surrounds the hole **21**A of the carrier **21** can be measured along the entire circumference thereof.
(37) Carrier Measurement Method
(38) Next, an example of a carrier measurement method using the carrier measuring device **1** having the above-described structure is described. In the exemplary embodiment, the upper thickness sensor **41** and the lower thickness sensor **42** each remain fixed at the middle in the longitudinal direction of the sensor frame **14**.
(39) (1) Measurement of Carrier **21** in Y Direction
(40) When the straight line L passing through the center C**2** of the carrier receiver **23**A and the center C**1** of the rotary table **23** is parallel to the Y direction as shown in FIG. **1**, the stage **12** in which the carrier **21** is placed is slid in the Y positive direction and concurrently a thickness of the carrier **21** is measured by the upper thickness sensor **41** and the lower thickness sensor **42**.
(41) A measurement pitch at this time can be set to any pitch and is, for instance, 0.2 mm. In this case, the measurement is performed at five points in a 1-mm section. A specific measurement position is on the straight line L (shown in FIG. **1**) passing through the center C**2** of the carrier receiver **23**A and the center C**1** of the rotary table **23** and in a range of 50 mm in the Y negative

direction from a position that is, for instance, approximately 5 to 7 mm distant from a point P.sub.1 of an edge (hole edge) **21**D defining the hole **21**A.
(42) (2) Measurement of Hole Surrounding Portion **21**C
(43) After the stage **12** is slid in the Y negative direction so that the hole surrounding portion **21**C that surrounds the hole **21**A formed in the carrier **21** is positioned between the upper thickness sensor **41** and the lower thickness sensor **42**, the stage **12** is stopped. In this state, the rotary table **23** is rotated and concurrently the thickness of the hole surrounding portion **21**C is measured by the upper thickness sensor **41** and the lower thickness sensor **42**. At this time, a measurement pitch in a rotation direction can be set to any pitch. For instance, the hole surrounding portion **21**C is measured at a total of 360 points (i.e., at one point per degree of the central angle of the hole **21**A) along the entire circumference thereof.
(44) A roll-off amount of the hole surrounding portion **21**C is defined as a value obtained by subtracting, from an average thickness of the carrier **21** in a reference surface defined as a reference for measuring the roll-off amount, a minimum thickness of the carrier **21** in a region of, for instance, 2 mm from the hole edge **21**D defining the hole **21**A toward the outer circumference of the carrier **21**. Here, the reference surface refers to a region of an upper surface of the carrier **21**, which is located at a predetermined distance from the hole edge **21**D toward the outer circumference of the carrier **21** and where the thickness of the hole surrounding portion **21**C is unlikely to change.
(45) FIG. **4**A is a top view of the carrier **21**. FIG. **4**B illustrates an enlarged view of a part **21**E of the hole surrounding portion **21**C and also illustrates a graph showing an example of a relationship between a distance on one line from the hole edge **21**D and the thickness of the carrier **21**.
(46) As is clear from the graph shown in FIG. **4**B, the thickness of the hole surrounding portion **21**C changes in a region located from the hole edge **21**D to a position slightly beyond 2 mm toward the outer circumference of the carrier **21**, but does not change outside the region located from the hole edge **21**D to the position slightly beyond 2 mm toward the outer circumference of the carrier **21**. A region of the upper surface of the carrier **21** that is approximately 3 mm to 5 mm distant from the hole edge **21**D toward the outer circumference of the carrier **21** can be thus defined as the reference surface. In FIG. **4**A, an annular region surrounded by two dashed lines is the reference surface. A measurement pitch for measuring the thickness of the hole surrounding portion **21**C from the hole edge **21**D toward the outer circumference of the carrier **21** can be set to any pitch and is, for instance, 0.2 mm. In this case, the measurement is performed at five points in a 1-mm section.
(47) Meanwhile, since the thickness of the hole surrounding portion **21**C significantly decreases in the region reaching 2 mm from the hole edge **21**D toward the outer circumference of the carrier **21**, a difference between the minimum thickness of the carrier **21** in this region and the average thickness of the carrier **21** in the reference surface is defined as the above-described roll-off amount on one line.
(48) Carrier Management Method
(49) Next, an example of a carrier management method using the carrier measuring device **1** and the carrier measurement method is described.
(50) (A) Carrier Management Method for Standard Semiconductor Wafers
(51) First, description is made on the management method of carriers **21** used in double-side polishing of standard semiconductor wafers that require a standard specification (e.g., flatness and surface roughness). In this carrier management method, only the measurement (1) of the carrier measurement method is performed on new carriers **21**.
(52) FIG. **5** is a flowchart showing an example of the carrier management method for the standard semiconductor wafers, the carrier management method being executed by the controller **51** of the computer **43**.
(53) In an initial state of the carrier measuring device **1**, the stage **12** is first positioned close to a

rear end of the platform **11**. In this initial state, the operator or a transfer mechanism (not shown) places each carrier **21** in the stage **12**. The carrier **21** is placed in the stage **12** in a direction where the center C**4** of the hole **21**A formed in the carrier **21** coincides with the center C**1** of the rotary table **23** as shown in FIG. **1**.
(54) The controller **51** of the computer **43** judges whether the carrier **21** is placed in a predetermined position (Step S**1**). This process is performed by, for instance, judging whether a detection signal from a sensor configured to detect a position of the stage **12** and a detection signal from a sensor configured to detect a housed state of the carrier **21** in the carrier receiver **23**A formed in the stage **12** are properly input.
(55) When a judgment result in Step S**1** is "NO", the controller **51** repeats the judgment process. When each of the detection signals is properly input, the judgment result in Step S**1** is "YES" and the controller **51** proceeds to Step S**2**.
(56) In Step S**2**, the controller **51** executes the measurement process of the carrier **21** in the Y direction, which is described in the measurement (1) of the carrier measurement method. Subsequently, the controller **51** proceeds to Step S**3**.
(57) In Step S**2**, the controller **51** controls the Y-axis motor **34** to slide the stage **12** in the Y positive direction and controls the upper thickness sensor **41** and the lower thickness sensor **42** to measure the thickness of the carrier **21** in a range of, for instance, 50 mm in the Y negative direction from the position that is approximately 5 to 7 mm distant from the point P.sub.1 (shown in FIG. **1**) of the hole **21**A, thereby obtaining a detection signal of the measurement results of the thickness.
(58) The controller **51** stores, in a predetermined storage area of the storage **52**, the measurement results of the thickness of the carrier **21** close to the point P.sub.1 together with an identification number assigned to the carrier **21**.
(59) In Step S**3**, the controller **51** judges whether all the carriers **21** are measured for the thickness in the Y direction. When the judgment result in Step S**3** is "NO", the controller **51** returns to Step S**1** and repeats the processes of Steps S**1** and S**2**.
(60) When the judgment result in Step S**3** is "YES" (i.e., when all the carriers **21** are measured for the thickness in the Y direction), the controller **51** proceeds to Step S**4**.
(61) In Step S**4**, the controller **51** calculates an average value of the measurement results of each carrier **21**, which are stored in the storage **52**, and classifies, on the basis of the obtained average values, the carriers **21** falling within a predetermined thickness range (for instance, in every 0.3-μm range) into the same class. Specifically, examples of the classes into which the carriers **21** are classified in terms of the average thickness include a class of 778.6 mm or less, a class in a range from 778.7 μm to 778.9 μm, and a class in a range from 783.5 μm to 783.7 μm.
(62) Next, the controller **51** selects a predetermined number (e.g., five) of carriers **21** from a plurality of carriers **21** belonging to the same class, as a set (carrier set) used together in the same batch in double-side polishing of a plurality of semiconductor wafers, and stores, in a predetermined storage area of the storage **52**, an identification number assigned to the carrier set. Subsequently, the controller **51** ends a series of processes.
(63) (B) Carrier Management Method for High-Precision Semiconductor Wafers
(64) Next, description is made on a management method of carriers **21** used in double-side polishing of high-precision semiconductor wafers that require a high-precision specification (e.g., flatness and surface roughness). In this carrier management method, both the measurements (1) and (2) of the carrier measurement method are performed.
(65) It should be noted that the processes of classifying a plurality of carriers **21** on the basis of average thicknesses thereof and subsequently forming a carrier set(s) are the same as those in the carrier management method (A) (see FIG. **5**), which is described in in paragraphs 0060 to 0070, and thus are briefly described below.
(66) When each carrier **21** is placed in a predetermined position of the stage **12** in a predetermined direction, the controller **51** controls the Y-axis motor **34**, the upper thickness sensor **41** and the

lower thickness sensor **42** to operate for measuring the thickness of the carrier **21** close to the point P.sub.1 of the hole **21**A. Subsequently, the controller **51** stores, in a predetermined storage area of the storage **52**, the measurement results together with an identification number of the carrier **21**.
(67) The controller **51** executes the above-described processes on all the carriers **21**. Subsequently, the controller **51** classifies each carrier **21** on the basis of the measurement results stored in the storage **52**, determines a carrier set including a plurality of carriers **21** belonging to the same class, and stores, in a predetermined storage area of the storage **52**, an identification number assigned to the carrier set.
(68) FIG. **6** is a flowchart showing an example of the carrier management method for the high-precision semiconductor wafers, the carrier management method being executed by the controller **51** of the computer **43**. Processes of Steps S**11** and S**12** are substantially the same as the processes of Steps S**1** and S**2**, respectively, which are shown in FIG. **5** and described in paragraphs 0060 to 0066, and thus are briefly described below.
(69) When each carrier **21** is placed in a predetermined position of the stage **12** in a predetermined direction (Step S**11**), the controller **51** controls the Y-axis motor **34**, the upper thickness sensor **41** and the lower thickness sensor **42** to operate for measuring the thickness of the carrier **21** close to the point P.sub.1 of the hole **21**A. Subsequently, the controller **51** stores, in a predetermined storage area of the storage **52**, the measurement results together with the identification number of the carrier **21** and the identification number of the carrier set to which the carrier **21** belongs (Step S**12**).
(70) In Step S**13**, the controller **51** executes the measurement process of the hole surrounding portion **21**C, which is described in the measurement (2) of the carrier measurement method. Subsequently, the controller **51** proceeds to Step S**14**.
(71) In Step S**13**, the controller **51** controls the Y-axis motor **34** to slide the stage **12** in the Y negative direction to position the hole surrounding portion **21**C between the upper thickness sensor **41** and the lower thickness sensor **42**. Next, the controller **51**, while the stage **12** is stopped, controls the table drive motor **24** to rotate the rotary table **23** and concurrently controls the upper thickness sensor **41** and the lower thickness sensor **42** to measure a roll-off amount (unit: μm) of the hole surrounding portion **21**C, and obtains a detection signal of the roll-off amount. In this case, the controller **51** controls the Y-axis motor **34**, the upper thickness sensor **41** and the lower thickness sensor **42** to operate for measuring the roll-off amount (unit: μm) of the hole surrounding portion **21**C, for instance, on a total of 360 lines (i.e., on one line per degree of the central angle of the hole **21**A) along the entire circumference of the hole surrounding portion **21**C, and obtains a detection signal of the roll-off amount.
(72) The controller **51** stores, in a predetermined storage area of the storage **52**, the measurement results of the roll-off amount of the carrier **21** along the entire circumference of the hole surrounding portion **21**C together with the identification number of the carrier **21** and the identification number of the carrier set to which the carrier **21** belongs.
(73) In Step S**14**, the controller **51** judges whether all carriers **21** belonging to the carrier set are measured for the thickness in the Y direction and the roll-off amount of the hole surrounding portion **21**C. When the judgment result in Step S**14** is “NO”, the controller **51** returns to Step S**11** and repeats the processes of Steps S**11** and S**13**.
(74) When the judgment result in Step S**14** is “YES” (i.e., when all the carriers **21** belonging to the carrier set are measured for the thickness in the Y direction and the roll-off amount of the hole surrounding portion **21**C), the controller **51** proceeds to Step S**15**.
(75) In Step S**15**, the controller **51** classifies the carriers into the carrier sets. Subsequently, the controller **51** ends a series of processes.
(76) Next, ranking of the carrier sets each including the plurality of carriers **21** is described. Table 1 shows an example of a rank table used for ranking the carrier sets.
(77) TABLE-US-00001 TABLE 1 Maximum Roll-off Amount of Hole Surrounding Portions Small

Medium Large Rank of Carrier Set A B C

(78) In a column showing items in Table 1, a "maximum roll-off amount of the hole surrounding portions" refers to a maximum value of roll-off amounts in the carrier set which are measured at 360 points (i.e., on 360 lines) of the hole surrounding portion **21**C that surrounds the hole **21**A in each carrier **21** by the measurement (2) of the carrier measurement method.

(79) The carrier sets are each ranked using this "maximum roll-off amount of the hole surrounding portions". Ranks may be appropriately set according to the "maximum roll-off amount of the hole surrounding portions". For instance, as shown in Table 1, the "maximum roll-off amounts of the hole surrounding portions" are classified into three groups, i.e., "Small", "Medium" and "Large", and the carrier sets are each ranked into any of "A", "B" and "C", which are set according to the above classification. A specification (mainly flatness) satisfied by the semiconductor wafers decreases in accordance with the ranks of the carrier set to be used, in order of A, B and C.

(80) Next, description is made on a relationship between the maximum roll-off amount of the hole surrounding portions and a roll-off amount of wafer outer circumferential portions in double-side polishing of semiconductor wafers using the carrier sets managed in the carrier management method. The following conditions are applied. (i) A double-side polishing device is mounted with a carrier set of five carriers **21** respectively configured to hold five semiconductor wafers and performs double-side polishing of the five semiconductor wafers. (ii) The double-side polishing device has a typical planetary gear mechanism. (iii) A total of 50 batches in the double-side polishing process are executed using different carrier sets. (iv) WaferSight 2 manufactured by KLA-Tencor Corporation is used as a measuring device configured to measure the roll-off amount of the wafer outer circumferential portion and the like.

(81) FIG. **7** is a diagram showing an example of a relationship, in each of the batches, between the maximum roll-off amount of the hole surrounding portions **21**C of the carriers **21** in the carrier set and the maximum roll-off amount of the wafer outer circumferential portions of the semiconductor wafers. In this example, the carrier set is used in performing double-side polishing of the semiconductor wafers according to the above conditions and the semiconductor wafers are obtained by performing double-side polishing in the batch. A method of ranking the carrier sets according to the maximum roll-off amount of the hole surrounding portions, which are shown in an abscissa axis, is the same as the method described regarding Table 1. The roll-off amount of the wafer outer circumferential portion, which is shown in an ordinate axis, is determined as Edge Site flatness Front reference least sQuare Deviation (ESFQD) that is one of indices for the flatness of the wafer outer circumferential portion of the semiconductor wafer.

(82) ESFQD is an unsigned maximum displacement amount from an in-site plane serving as a reference plane. The in-site plane is determined by defining fan-shaped sectors in the wafer outer circumferential portion of the semiconductor wafer and calculating height data in each sector by a method of least squares. Each site in the sector has one value of ESFQD.

(83) It is clear from FIG. **7** that the larger the maximum roll-off amount of the hole surrounding portions **21**C of the carriers **21** is, the larger the roll-off amount of the wafer outer circumferential portions is. A, B and C shown in FIG. **7** respectively correspond to the ranks A, B and C of the carrier set, which are shown in Table 1.

(84) As described above, in the exemplary embodiment, since the thickness of the hole surrounding portion **21**C of each carrier **21** can be measured along the entire circumference of the hole surrounding portion **21**C by the measurement (2) of the carrier measurement method, a local roll-off of the hole surrounding portion **21**C, which is conventionally not detectable, can be detected.

(85) Further, in the exemplary embodiment, since the upper thickness sensor **41** and the lower thickness sensor **42** each remain fixed at the middle in the longitudinal direction of the sensor frame **14**, the thickness and a shape of the carrier **21** can be highly accurately measured and, unlike the conventional art where thickness sensors are positioned by being rotated, positioning control of the upper thickness sensor **41** and the lower thickness sensor **42** is eliminated.

(86) Furthermore, in the exemplary embodiment, the thickness and shape of the plurality of carriers **21** forming each carrier set are measured by the measurements (1) and (2) of the carrier measurement method and the carrier set is ranked in advance according to the rank table shown in Table 1 using the measurement results. Then, the carrier set in the rank corresponding to the specification required for the semiconductor wafers is used to perform double-side polishing of the semiconductor wafers. Thus, the semiconductor wafers satisfying the required specification can be obtained.
(87) It should be noted that the specific arrangement of the invention, which is not limited by the exemplary embodiments of the invention described in detail with reference to the attached drawings, encompasses design modifications and the like as long as such modifications are compatible with an object of the invention.
(88) For instance, in the above exemplary embodiment, the roll-off amount of the hole surrounding portion **21**C is defined as a value obtained by subtracting, from an average thickness of the carrier **21** in the reference surface, a minimum thickness of the carrier **21** in a region of 2 mm from the hole edge **21**D defining the hole **21**A toward the outer circumference of the carrier **21**. However, the roll-off amount of the hole surrounding portion **21**C is not necessarily defined in this manner. For instance, regarding the carrier **21** for the high-precision semiconductor wafer, the roll-off amount of the hole surrounding portion **21**C may be defined as a value obtained by subtracting, from an average thickness of the carrier **21** in the reference surface, a minimum thickness of the carrier **21** in a region of 1 mm from the hole edge **21**D defining the hole **21**A toward the outer circumference of the carrier **21**.
(89) The above exemplary embodiment shows the example in which the slide unit **13** configured to slide the stage **12** in the Y direction includes the guide rails **32**, the ball screw **33**, the Y-axis motor **34** and the like. Alternatively, the slide unit **13** may include a linear motor and the like.
(90) The above exemplary embodiment shows the example in which the upper thickness sensor **41** and the lower thickness sensor **42** are each a laser displacement sensor. Alternatively, the upper thickness sensor **41** and the lower thickness sensor **42** may be each any other type of non-contact sensor such as a capacitance sensor.
(91) The above exemplary embodiment shows the example in which the stage **12** is slid by the slide unit **13** in the Y direction while the upper thickness sensor **41** and the lower thickness sensor **42** are each fixed to the sensor frame **14**. Alternatively, the sensor frame **14** to which the upper thickness sensor **41** and the lower thickness sensor **42** are fixed may be slid in the Y direction while the stage **12** is fixed. A slide unit configured to slide the sensor frame **14** in this arrangement may be configured in the same manner as the above-described slide unit **13**. Additionally, both the stage **12** and the sensor frame **14** may be slid in the Y direction.

Claims

1. A carrier measuring device configured to measure a thickness of a carrier having a hole in which a disc-shaped semiconductor wafer is to be held, the hole being eccentric with respect to the carrier, the carrier measuring device comprising: a rotary table comprising a carrier receiver configured to horizontally house the carrier; a table drive motor configured to rotate the rotary table around a center axis thereof as a rotation axis; an upper thickness sensor and a lower thickness sensor configured to be positioned above and below the carrier, respectively, and measure the thickness of the carrier in a non-contact manner; and a slide unit configured to slide, in a horizontal direction, one or both of: the rotary table; and the upper thickness sensor and the lower thickness sensor, wherein the carrier receiver is formed to be capable of housing the carrier in a manner that a center of the hole of the carrier coincides with a center of the rotary table.
2. The carrier measuring device according to claim 1, further comprising a sensor frame attached with the upper thickness sensor and the lower thickness sensor between which the rotary table is

capable of passing through in the horizontal direction.

3. The carrier measuring device according to claim 2, wherein the upper thickness sensor and the lower thickness sensor are attached to the sensor frame in a manner to be respectively positioned above and below a straight line passing through the center of the rotary table and a center of the carrier receiver, and the slide unit slides one or both of the rotary table and the sensor frame in the horizontal direction parallel to the straight line.

4. A carrier measurement method for measuring, by using the carrier measuring device according to claim 1, a thickness of a disc-shaped carrier having a hole in which a disc-shaped semiconductor wafer is to be held, the hole being eccentric with respect to the carrier, the carrier measurement method comprising: first measuring the thickness of the carrier at a predetermined point while sliding, in the horizontal direction, one or both of: the rotary table in which the carrier is housed; and the upper thickness sensor and the lower thickness sensor; and second measuring a thickness of a hole surrounding portion that surrounds the hole of the carrier while rotating the rotary table in which the carrier is housed.

5. A carrier management method for managing, on a basis of measurement results obtained through measurement using the carrier measuring device according to claim 1, a plurality of disc-shaped carriers each having a hole in which a disc-shaped semiconductor wafer is to be held, the hole being eccentric with respect to the corresponding carrier, the carrier management method comprising: first measuring the thickness of each of the carriers at a predetermined point while sliding, in the horizontal direction, one or both of: the rotary table in which the carrier is housed; and the upper thickness sensor and the lower thickness sensor; classifying the carriers falling within a predetermined range of the thickness into the same class on a basis of the measurement results in the first measuring; selecting a predetermined number of carriers from the carriers belonging to the same class to determine a carrier set to be used in the same batch in double-side polishing of the semiconductor wafers; second measuring a thickness of a hole surrounding portion that surrounds the hole of each of the carriers of the carrier set while rotating the rotary table in which the carriers are housed; and ranking the carrier set on a basis of the measurement results in the first and second measuring according to a specification required for the semiconductor wafers.

6. The carrier management method according to claim 5, wherein in the ranking, a roll-off amount of the hole surrounding portion of each of the carriers is calculated on a basis of the measurement results in the second measuring and the carrier set is ranked on a basis of the measurement results in the first measuring and the roll-off amounts of the hole surrounding portions.